

COMICCRAFT – DEPLOYMENT DOCUMENTATION

1. Introduction

Deployment is the process of preparing the ComicCraft application so that it can be run and accessed by users.

This document describes the deployment requirements, setup process, configuration, and execution of the ComicCraft AI Comic Story Creator application.

2. Deployment Objective

The main objectives of deployment are:

- Prepare the application for execution
- Configure required software and dependencies
- Configure API credentials securely
- Start the FastAPI backend
- Make the web interface accessible
- Verify the application after deployment

3. Deployment Requirements

The following requirements are needed to run ComicCraft:

Hardware

- Computer or laptop
- Minimum 4 GB RAM
- Sufficient storage space
- Stable internet connection

Software

- Windows, Linux, or macOS
- Python 3.x
- Visual Studio Code or another code editor
- Modern web browser
- Required Python packages
- Gemini API access

4. Project Structure

The ComicCraft application contains major components such as:

```
ComicCraft/
|
├─ app/
├─ templates/
├─ static/
|   └─ panels/
|       └─ exports/
├─ requirements.txt
└─ configuration files
```

The exact structure may vary depending on the final implementation.

5. Environment Setup

The application can be deployed in a Python virtual environment.

The basic setup process is:

1. Open the ComicCraft project folder.
2. Open the terminal.
3. Create a Python virtual environment.
4. Activate the virtual environment.
5. Install the required dependencies.
6. Configure the required API credentials.
7. Start the FastAPI application.
8. Open the application in a web browser.

6. Dependency Installation

The required Python packages are listed in the project's `requirements.txt` file.

The dependencies should be installed before running the application.

The installation process ensures that the required frameworks and libraries are available to the application.

7. API Configuration

ComicCraft requires access to the Gemini AI service.

The API key should be configured securely using the project's configuration or environment variables.

The API key should not be:

- Hard-coded into source files

- Uploaded to GitHub
- Shared publicly
- Included in screenshots

Sensitive configuration information should remain private.

8. Running the Application

After completing the environment setup, the FastAPI application can be started using the project's configured startup command.

Once the server starts successfully, the application can be opened through a web browser using the local server address provided by the application.

9. Deployment Verification

After starting the application, the following items should be verified:

- Home page loads correctly
- Story prompt can be entered
- Prompt validation works
- Gemini AI responds correctly
- Comic panels are generated
- Images are processed
- Comic layout is created
- PDF generation works
- Generated files can be accessed

10. Deployment Workflow

The deployment workflow is:

Project Files

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Python Environment

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Install Dependencies

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Configure API Credentials

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Start FastAPI Server

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Open Web Application

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Test Application

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ComicCraft Ready for Use

11. Deployment Considerations

The following points should be considered during deployment:

- Keep API credentials secure.
- Install compatible dependency versions.
- Maintain correct project folder paths.
- Ensure required directories exist.
- Maintain a stable internet connection for AI services.
- Check generated image and PDF file locations.
- Handle application errors properly.

12. Local Deployment

During development and demonstration, ComicCraft can be run locally on a computer.

Local deployment is useful for:

- Development
- Testing
- Demonstration
- Debugging
- Project evaluation

The application can be accessed through a web browser after the local FastAPI server is started.

13. Future Deployment

The ComicCraft application can be deployed to a cloud or web hosting platform in the future.

Possible improvements include:

- Cloud hosting
- Production database
- Secure secret management
- Scalable image generation
- User authentication
- Cloud file storage
- Monitoring and logging

14. Expected Result

After successful deployment, the ComicCraft application should be accessible through its web interface.

Users should be able to enter a story idea, generate a comic using AI, view the generated panels, and create the final PDF output.

15. Conclusion

The Deployment Phase prepares ComicCraft for execution and demonstration.

Proper environment setup, dependency installation, secure API configuration, application startup, and deployment verification ensure that the ComicCraft application is ready for use.